

RONALD YOU

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EDUCATION

University of Waterloo – Bachelor of Software Engineering, Honours
3.7/4.0 GPA (87% cumulative average)

4th Year

EXPERIENCE

Waabi Ai

Research Engineer

May 2026 - Present

Toronto, ON

- Spearheaded a new audio perception modality for autonomous trucks, training a data-efficient machine learning model to detect sirens, which increased active emergency vehicle (EMV) recognition by **30 percentage points**
- Productionized the end-to-end audio perception pipeline in Rust, integrating sensor ingestion, preprocessing, inference, and detection outputs across the onboard autonomy stack
- Optimized real-time audio processing through parallelization, achieving **3–5 ms** end-to-end detection latency with **0 ms** additional critical-path latency
- Built critical downstream integrations, including real-time alerts for the Remote Assist team and automated data tagging systems to streamline future dataset curation

Palona AI

Software Engineer

Jan 2026 - Apr 2026

Los Altos, CA

- Architecting a real-time voice agent backend in Python, optimizing LLM inference pipelines to minimize time-to-first-token (TTFT) and ensure sub-second conversational latency
- Pioneered an automated agent evaluation framework utilizing LLM-generated test cases, accelerating the QA process and reducing manual testing time by **50%**
- Engineered a dynamic latency-masking algorithm that injects context-aware filler words during processing gaps, smoothing audio transitions and improving user retention

Arcturus Networks Inc.

Machine Learning Engineer

May 2025 - Aug 2025

Toronto, ON

- Devised a full-stack data verification loop by integrating **CVAT** and **FiftyOne** into the backend pipeline, enabling a **Human-In-The-Loop** workflow that streamlined dataset validation
- Designed and orchestrated a data drift detection system using **Apache Airflow**, scheduling statistical checks on production inputs to automatically flag feature divergence
- Implemented custom weighted divergence algorithms within the Airflow DAGs, achieving 90% precision in identifying shifts correlated with model performance degradation

Martinrea International Inc.

Computer Vision Software Engineer

Sept 2024 – Dec 2024

Toronto, ON

- Developed and implemented deep learning models for in-house quality inspection systems using **PyTorch** and **OpenCV**, saving **\$30K** per installation compared to external solutions
- Utilized **YOLO v8** object detection and classification models with **Ultralytics**, **NumPy**, and **Matplotlib** to train custom presence detection models obtaining **99%** accuracy and precision
- Reduced cycle time by **100x** (3s to 30 ms) by re-architecting the pipeline: downsampled capture resolution for latency, deployed TensorRT models, and optimized edge hardware power modes

PROJECTS

Adaptive Cruise Control Simulator | C++, CMake, OpenGL, GitHub

Present

- Engineered a real-time C++ simulation engine with an OpenGL rendering pipeline to visualize vehicle kinematics and complex multi-agent interactions at 60+ FPS
- Developed a "semi-AI" traffic ecosystem using Finite State Machines, enabling NPC vehicles to utilize spatial awareness for autonomous lane changes, collision avoidance, and speed matching
- Implemented a PID controller with anti-windup and actuator saturation, ensuring system stability while navigating stochastically generated terrain profiles (inclines, friction patches)

SKILLS

Languages: Python, C++, Go, TypeScript, SQL, Java, C

Frameworks & Libraries: React, Node.js, Prisma, PyTorch, OpenCV, Ultralytics, NumPy, PostgreSQL

Dev Tools & Platforms: AWS, Docker, Apache Airflow, Linux, Git, CUDA, TensorRT, CVAT